

02

Build Your Pong Game



Lingma



1. Open VSCODE and click on "Extensions" on the left side.
2. Enter "Lingma" in the search box, click the first option, and then click the Install button.
3. After installation, a "Lingma" button will appear under the "Extensions" button on the left navigation bar. Click to open the plugin.

The Classic Pong Game



Home Pong quickly soared to popularity in the mid-1970s
(Image credit: Interfoto/Warry Stock Photo)

Generate a classic Pong game,
including Player battles and
Player vs PC

Review of The Codes

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<head>
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  <title>Pong Game</title>
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</head>
<body>
  // divisions in the page
  <script>
    // javascript codes
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// This is a single-line comment

/*

This is a multi-line comment

*/

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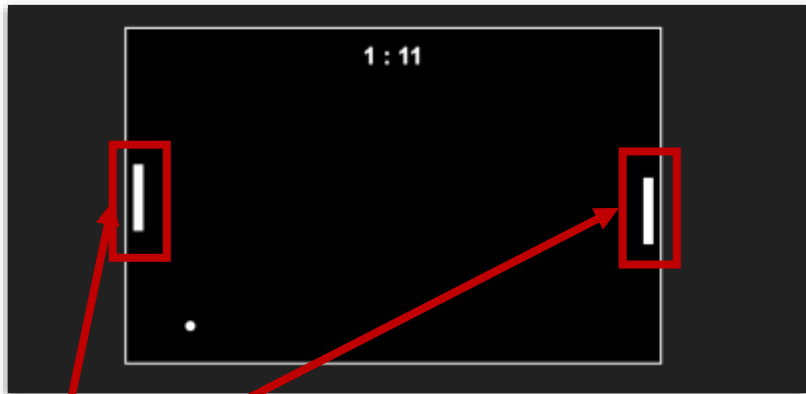
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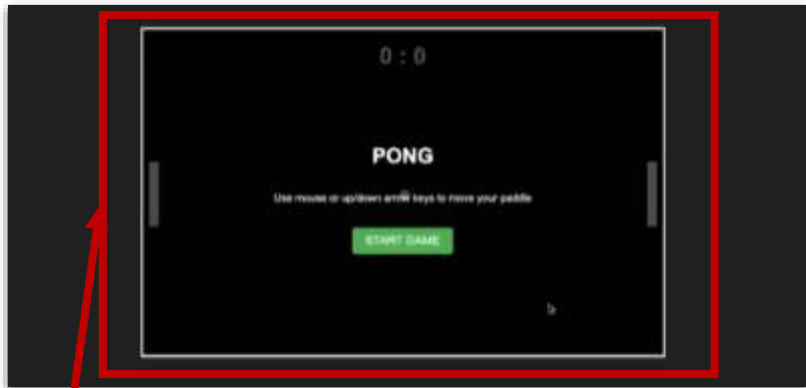
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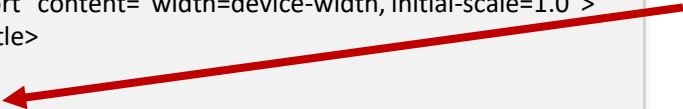
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CSS Style Codes

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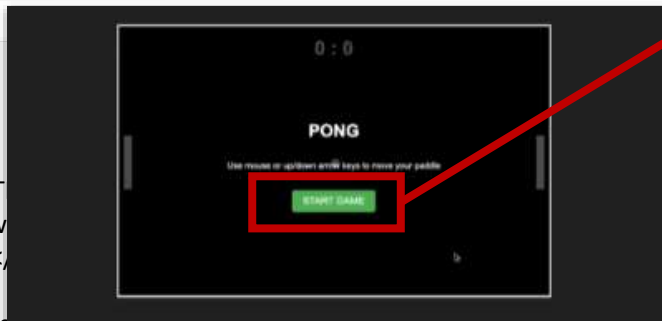


```
#start-screen {
  position: absolute;
  width: 100%;
  height: 100%;
  display: flex;
  flex-direction: column;
  justify-content: center;
  align-items: center;
  background-color: rgba(0, 0, 0, 0.7);
  color: #fff;
  z-index: 10;
}
```

```
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JavaScript Codes

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```



```
// Event listeners
startButton.addEventListener('click', function() {
  startScreen.style.display = 'none';
  playerScore = 0;
  computerScore = 0;
  isGameRunning = true;
  initGame();
  animationId = requestAnimationFrame(gameLoop);
});
```

```
background-color: rgba(0, 0, 0, 0.7);
color: #fff;
z-index: 10;
}
```

```
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Can AI Explain The Codes?

Explain how does the computer control the paddle to catch the ball.

2dpong.html

Explain how does the computer control the paddle to catch the ball.

Lingma

I'll explain how the computer controls the paddle to catch the ball in this Pong game.

Looking at the code, the AI control logic is implemented in the `aiControl()` function (lines 258-280). Here's how it works:

AI Control Strategy

The computer uses a **reactive tracking system** with these key behaviors:

1. Ball Direction Detection

```
javascript
if (ball.dx > 0) {
    // Ball is moving towards the AI paddle (right side)
} else {
    // Ball is moving away from the AI paddle
}
```

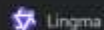
The AI only actively tracks the ball when it's moving towards the right paddle (`ball.dx > 0`).

```
2dpong.html > html > body > script > aiControl
2
<html lang="en">
100
<body>
124
  <script>
240
    function handleInput() {
258
    }
259
    } else {
260
        // AI for PC
261
        aiControl();
262
    }
263
    }
264
265
    function aiControl() {
266
        const paddleCenter = player2.y + player2.height / 2;
267
        const ballCenter = ball.y;
268
        const reactionSpeed = player2.speed * 0.7; // slightly slower than player
269
270
        // Only move if ball is moving towards AI
271
        if (ball.dx > 0) {
272
            if (paddleCenter < ballCenter - 10) {
273
                player2.dy = reactionSpeed;
274
            } else if (paddleCenter > ballCenter + 10) {
275
                player2.dy = -reactionSpeed;
276
            } else {
277
                player2.dy = 0;
278
            }
279
        } else {
280
            // Return to center when ball is moving away
281
            const centerY = canvas.height / 2;
282
            if (paddleCenter < centerY - 20) {
```

Ask AI for Some Suggestions

Suggest some strategies for the player to help them win against the computer.

Suggest some strategies for the player to help them win against the computer, but don't change the code.



Based on the AI control logic I analyzed, here are effective strategies to beat the computer opponent:

Winning Strategies Against the AI

1. Use Angled Shots

The AI only tracks the ball's current position—it doesn't predict trajectory. When you hit the ball with the **edges of your paddle**, it creates sharp angles:

javascript

```
// From the code (Lines 307-310):  
const hitPos = (ball.y - player1.y) / player1.height;  
ball.dy = 8 * (hitPos - 0.5);
```

Strategy: Hit the ball with the top or bottom third of your paddle to create steep diagonal shots that change direction quickly, making it harder for the AI to react in time.

2. Exploit the Reaction Speed Limitation

The AI moves at **70% of your speed** (`reactionSpeed = player2.speed * 0.7`). This means:

Strategy: Make rapid directional changes by hitting the ball back and forth quickly. The AI will lag behind because it's slower than you!

3. Target the Corners

Since the AI has a **10-pixel dead zone** before it reacts, very fast balls moving toward the top or bottom corners can slip past:

Strategy:

- Position yourself near the top or bottom of the court
- Hit the ball when it's already moving diagonally
- This maximizes the angle and speed toward the corners

Ask AI for Some Suggestions

4. Watch for the "Return to Center" Weakness

When the ball moves away from the AI (`ball.dx < 0`), it slowly returns to center at **half speed**:

Strategy:

- After the AI misses or you score, immediately prepare for a quick return shot
- The AI might still be repositioning when you hit the ball back
- Catch it off-guard during this vulnerable moment

5. Increase Ball Speed Strategically

Each paddle hit increases ball speed by 0.5 (up to max 12):

javascript



```
ball.speed = Math.min(ball.speed + 0.5, 12);
```

Strategy:

- Engage in longer rallies to build up speed
- Once the ball is fast (>9-10 speed), the AI's 70% reaction speed becomes a bigger disadvantage
- Fast balls give the AI less time to reach extreme positions

Suggest some strategies for the player to help them win against the computer.

6. Avoid Predictable Patterns

The AI will eventually track consistent patterns.

Strategy:

- Vary your paddle positions (don't always stay in the center)
- Mix up when you use angled vs. straight shots
- Occasionally let the ball pass to reset its speed if it gets too fast for you

7. Defensive Positioning

Keep your paddle near the **middle third** of the court most of the time:

Strategy:

- This gives you maximum flexibility to reach both high and low shots
- Only move to extremes when you're setting up an angled attack
- Stay centered to defend against the AI's shots while planning your counterattacks

Ask AI for Some Suggestions

Suggest some strategies for the player to help them win against the computer.

8. Patience Over Aggression

The AI makes mistakes when the ball speed is high and angles are sharp.

Strategy:

- Don't rush to win—let the AI make errors
- Focus on consistent returns rather than risky shots
- Wait for the perfect opportunity to strike with a well-placed angled shot

Pro Tip Combination

Best approach: Build up ball speed through 3-4 safe returns, then suddenly switch to an extreme angled shot using the edge of your paddle. The combination of high speed + sharp angle + AI's slower reaction = point for you! 🏆

Improve The Codes

Improve the code:

1. each player (left and right) should have an independent mode, either Human or PC.
2. In PC mode, the paddle is controlled by the program;
3. in Human mode, it is controlled by the player.
4. The left player uses W (up) and S (down), while the right player uses the Up and Down arrow keys.
5. Add on-screen hints for the controls, and include buttons to Start, Pause, and Restart the game.

Improve The Codes

Improve the code:

- 1.add randomly generated buff blocks that appear in the same column as the paddles.
- 2.When a paddle moves and touches a buff block, it gains a special effect for several rounds (one round is defined as the ball moving from one side to the other and back).

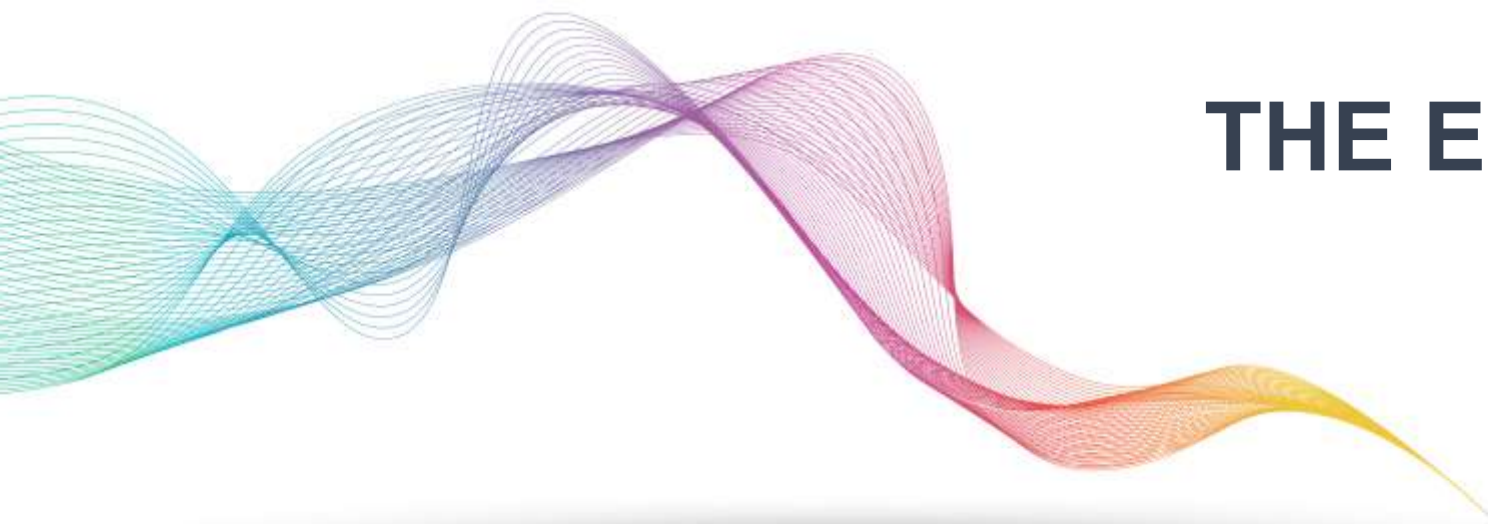
Examples of buff blocks include:

- Slow down block: slows the ball speed by 20% for 3 rounds .
- Split block: splits the ball into 3 copies; the game continues if the opposing player catches at least one copy, but each missed ball gives a point. No round limit .
- Color block: changes the ball's color randomly for 5 rounds.

Improve The Codes

Setting Page:

1. Move the buttons into an operation bar above the play area, and add a button to open a settings page (when the settings page is displayed, the game pauses automatically).
2. Add a slider in the settings page to select the buff refresh speed level: level 0 means no buffs are generated, level 1 means 1 block every 10 rounds, and level 5 means about 5 blocks every 10 rounds.
3. Add an integer field in the settings page to set the base speed of the ball (the starting speed).
4. Add fields to set minimum and maximum limits for the ball speed, and a field to set the speed change ratio, from 1.1 to 1.5.
5. Add fields to set the minimum and maximum paddle length, and a field to set the length change ratio, from 1.1 to 1.5.



THE END